1 Lecture 10 Interpretation of OLS Short Answer Solutions

2 Question #1

The OLS estimator is

$$b = (X'X)^{-1}X'Y,$$

where X is $n \times K$, Y is $n \times 1$, and b is $K \times 1$.

Define the prediction error associated with the OLS coefficients as

$$e_i = y_i - x_i'b.$$

Let H denote the $n \times 1$ vector of OLS prediction errors:

$$H = \begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{bmatrix}.$$

a. Define the projection matrix for X, P_X.

Solution:

The projection matrix for X is

$$\mathrm{P}_X = X(X'X)^{-1}X'.$$

b. Show that

$$\mathrm{P}_X H = 0.$$

Solution:

Since

$$H = Y - Xb,$$

we have

$$\begin{aligned} \mathrm{P}_X H &= X(X'X)^{-1}X'(Y - Xb) \\ &= X(X'X)^{-1}X'Y - X(X'X)^{-1}X'Xb. \end{aligned}$$

Using

$$b = (X'X)^{-1}X'Y,$$

it follows that

$$\mathrm{P}_X H = Xb - Xb = 0.$$

c. State the result from part (b) in words.

Solution:

The OLS prediction errors cannot be predicted by a linear combination of the columns of X.

d. Consider predicting the OLS errors using X:

$$\hat{e}_i = x_{i1}a_1 + x_{i2}a_2 + \cdots + x_{iK}a_K.$$

Show that the OLS coefficient vector a is equal to zero.

Solution:

Using H as the dependent variable, the OLS estimator is

$$a = (X'X)^{-1}X'H.$$

Since

$$H = Y - Xb,$$

we have

$$\begin{aligned} a &= (X'X)^{-1}X'(Y - Xb) \\ &= (X'X)^{-1}X'Y - (X'X)^{-1}X'Xb \\ &= b - b \\ &= 0. \end{aligned}$$

3 Question #2

Let Y, Z, X, and $\boldsymbol{\varepsilon}$ be $n \times 1$ vectors related by

$$Y = \tau Z + \kappa X + \boldsymbol{\varepsilon}.$$

Suppose every element of Z is equal to 1. For context, let Y denote total sales across n observations, X the price at each observation, and $\boldsymbol{\varepsilon}$ all other factors affecting sales.

a. Define the projection matrix for Z as

$$P_Z = Z(Z'Z)^{-1}Z'.$$

Describe P_Z completely. What are its dimensions, and what is the value of every element of the matrix?

Solution:

Since Z is an $n \times 1$ vector of ones,

$$Z'Z = \sum_{i=1}^{n} 1 \cdot 1 = n.$$

Therefore,

$$P_Z = Z(Z'Z)^{-1}Z' = \frac{1}{n}ZZ'.$$

Since

$$ZZ' = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & \cdots & 1 \end{bmatrix} = \mathbf{1}_{n \times n},$$

where $\mathbf{1}_{n \times n}$ is an $n \times n$ matrix of ones,

$$P_Z = \frac{1}{n} \mathbf{1}_{n \times n}.$$

Thus, P_Z is an $n \times n$ matrix in which every element is $1/n$.

b. Consider the matrix product $P_Z X$. Describe the result completely. What are its dimensions, and what is the value of every element of the resulting vector?

Solution:

Since P_Z is $n \times n$ and X is $n \times 1$, $P_Z X$ is $n \times 1$.

Since every element of P_Z is $1/n$,

$$P_Z X = \begin{bmatrix} \frac{1}{n} \sum_{i=1}^{n} x_i \\ \frac{1}{n} \sum_{i=1}^{n} x_i \\ \vdots \\ \frac{1}{n} \sum_{i=1}^{n} x_i \end{bmatrix} = \begin{bmatrix} \bar{x} \\ \bar{x} \\ \vdots \\ \bar{x} \end{bmatrix}.$$

Thus, every element of $P_Z X$ is the sample average of the observations in X.

c. Define the residual-maker matrix for Z as

$$M_Z = I - Z(Z'Z)^{-1}Z'.$$

Multiplying the model by M_Z gives

$$Y^* = \kappa X^* + \varepsilon^*,$$

Describe X^* in this equation completely. What are its dimensions, and what is the value of every element of the vector?

Solution:

Work this out on your own.